Sri Lanka Institute of Information Technology

IT2130 – Operating Systems and System Administration
Year 2, Semester 2- 2026

Practical Answer Submission

Practical Sheet No: 9

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Objective: This practical demonstrates the use of file management in Unix/Linux.

Activity 1

Unix directory structure

1. Log in to the system and type pwd to print working directory. Note down the absolute path, i.e. pathname starting from /, i.e. root directory.

```
/home/semini/.hushlogin file.
semini@Semini-Chamathka:~$ pwd
/home/semini
```

2. Now use cd command to change your directory to /. Type **ls -l** and observe the different fields that can be seen in the output.

   How can you recognize file/directory by looking at this output? Do you have the read, write, or execute permission to these files/directories? What do these fields mean?

```
semini@Semini-Chamathka:~$ cd /
ls -l
total 2796
lrwxrwxrwx   1 root root       7 Apr 22  2024 bin -> usr/bin
drwxr-xr-x   2 root root    4096 Feb 26  2024 bin.usr-is-merged
drwxr-xr-x   2 root root    4096 Apr 22  2024 boot
drwxr-xr-x  15 root root    3860 Mar 31 10:17 dev
drwxr-xr-x  88 root root    4096 Mar 31 10:17 etc
drwxr-xr-x   3 root root    4096 Jan 28 10:07 home
-rwxr-xr-x   1 root root 2781568 Dec 12 07:28 init
lrwxrwxrwx   1 root root       7 Apr 22  2024 lib -> usr/lib
drwxr-xr-x   2 root root    4096 Apr  8  2024 lib.usr-is-merged
lrwxrwxrwx   1 root root       9 Apr 22  2024 lib64 -> usr/lib64
drwx------   2 root root   16384 Jan 28 10:05 lost+found
drwxr-xr-x   2 root root    4096 Aug  5  2025 media
drwxr-xr-x   7 root root    4096 Jan 28 10:06 mnt
drwxr-xr-x   2 root root    4096 Aug  5  2025 opt
dr-xr-xr-x 336 root root       0 Mar 31 10:17 proc
drwx------   3 root root    4096 Aug  5  2025 root
drwxr-xr-x  19 root root     560 Mar 31 10:17 run
lrwxrwxrwx   1 root root       8 Apr 22  2024 sbin -> usr/sbin
drwxr-xr-x   2 root root    4096 Mar 31  2024 sbin.usr-is-merged
drwxr-xr-x   2 root root    4096 Jan 28 10:06 snap
drwxr-xr-x   2 root root    4096 Aug  5  2025 srv
dr-xr-xr-x  13 root root       0 Mar 31 10:16 sys
drwxrwxrwt   8 root root    4096 Mar 31 10:17 tmp
drwxr-xr-x  12 root root    4096 Aug  5  2025 usr
drwxr-xr-x  13 root root    4096 Jan 28 10:06 var
```

   Frist character shows the type:

   d- director

   - File

   l- symbolic link

   permission meaning

   r- read / w- write / x- execute

3. Visit following directories:

   /bin, /home, /dev, /etc, /lib

   Type ls -l under each directory. Can you interpret the output of ls -l command?

   Check the very first letter of each line when you type ls -l under /dev directory.

```
semini@Semini-Chamathka:/lib$ cd /bin && ls -l
total 132620
lrwxrwxrwx 1 root root           4 Feb 10  2024  NF -> col1
lrwxrwxrwx 1 root root           1 Apr  8  2024  X11 -> .
-rwxr-xr-x 1 root root       55744 Apr  5  2024 '['
-rwxr-xr-x 1 root root       18744 Mar 19  2025  aa-enabled
-rwxr-xr-x 1 root root       18744 Mar 19  2025  aa-exec
-rwxr-xr-x 1 root root       18736 Mar 19  2025  aa-features-abi
-rwxr-xr-x 1 root root       16422 Feb 19  2025  add-apt-repository
-rwxr-xr-x 1 root root       14720 Mar  6 21:30  addpart
lrwxrwxrwx 1 root root          26 Dec  3 20:31  addr2line -> x86_64-linux-g
nu-addr2line
-rwxr-xr-x 1 root root        2322 Apr 18  2024  apport-bug
-rwxr-xr-x 1 root root       13625 Jul  8  2025  apport-cli
lrwxrwxrwx 1 root root          10 Jul  8  2025  apport-collect -> apport-bu
g
-rwxr-xr-x 1 root root        3790 Jul  8  2025  apport-unpack
-rwxr-xr-x 1 root root      141544 Apr  8  2024  appstreamcli
lrwxrwxrwx 1 root root           6 Apr  8  2024  apropos -> whatis
-rwxr-xr-x 1 root root       18824 Oct 22  2024  apt
lrwxrwxrwx 1 root root          18 Feb 19  2025  apt-add-repository -> add-a
pt-repository
-rwxr-xr-x 1 root root       88544 Oct 22  2024  apt-cache
-rwxr-xr-x 1 root root       27104 Oct 22  2024  apt-cdrom
-rwxr-xr-x 1 root root       31120 Oct 22  2024  apt-config
-rwxr-xr-x 1 root root       23008 Oct 22  2024  apt-extracttemplates
-rwxr-xr-x 1 root root      227816 Oct 22  2024  apt-ftparchive
-rwxr-xr-x 1 root root       51680 Oct 22  2024  apt-get
-rwxr-xr-x 1 root root       28664 Oct 22  2024  apt-key
-rwxr-xr-x 1 root root       55776 Oct 22  2024  apt-mark
-rwxr-xr-x 1 root root       51608 Oct 22  2024  apt-sortpkgs
lrwxrwxrwx 1 root root          19 Dec  3 20:31  ar -> x86_64-linux-gnu-ar
-rwxr-xr-x 1 root root       35336 Apr  5  2024  arch
lrwxrwxrwx 1 root root          19 Dec  3 20:31  as -> x86_64-linux-gnu-as
-rwxr-xr-x 1 root root         984 Jan  2  2024  automat-visualize3
lrwxrwxrwx 1 root root          21 Apr  8  2024  awk -> /etc/alternatives/aw
k
-rwxr-xr-x 1 root root       55816 Apr  5  2024  b2sum
-rwxr-xr-x 1 root root       39432 Apr  5  2024  base32
-rwxr-xr-x 1 root root       39432 Apr  5  2024  base64
```

```
semini@Semini-Chamathka:/home$ cd /dev && ls -l
total 0
crw-r--r-- 1 root root     10, 235 Mar 31 10:17 autofs
drwxr-xr-x 2 root root         600 Mar 31 10:17 block
drwxr-xr-x 2 root root         120 Mar 31 10:17 bsg
crw------- 1 root root     10, 234 Mar 31 10:17 btrfs-control
drwxr-xr-x 2 root root        2860 Mar 31 10:17 char
crw------- 1 root root      5,   1 Mar 31 10:17 console
lrwxrwxrwx 1 root root          11 Mar 31 10:17 core -> /proc/kcore
crw------- 1 root root     10, 123 Mar 31 10:17 cpu_dma_latency
crw------- 1 root root     10, 203 Mar 31 10:17 cuse
drwxr-xr-x 6 root root         120 Mar 31 10:17 disk
crw-rw-rw- 1 root root     10, 125 Mar 31 10:17 dxg
lrwxrwxrwx 1 root root          13 Mar 31 10:17 fd -> /proc/self/fd
crw-rw-rw- 1 root root      1,   7 Mar 31 10:17 full
crw-rw-rw- 1 root root     10, 229 Mar 31 10:17 fuse
crw------- 1 root root     10, 228 Mar 31 10:17 hpet
drwxr-xr-x 2 root root           0 Mar 31 10:17 hugepages
crw------- 1 root root    229,   0 Mar 31 10:17 hvc0
crw------- 1 root root    229,   1 Mar 31 10:17 hvc1
crw--w---- 1 root tty     229,   2 Mar 31 10:17 hvc2
crw------- 1 root root    229,   3 Mar 31 10:17 hvc3
crw------- 1 root root    229,   4 Mar 31 10:17 hvc4
crw------- 1 root root    229,   5 Mar 31 10:17 hvc5
crw------- 1 root root    229,   6 Mar 31 10:17 hvc6
crw------- 1 root root    229,   7 Mar 31 10:17 hvc7
crw------- 1 root root     10, 183 Mar 31 10:17 hwrng
lrwxrwxrwx 1 root root          12 Mar 31 10:17 initctl -> /run/initctl
crw-r--r-- 1 root root      1,  11 Mar 31 10:17 kmsg
crw-rw---- 1 root kvm      10, 232 Mar 31 10:17 kvm
lrwxrwxrwx 1 root root          28 Mar 31 10:17 log -> /run/systemd/journa
dev-log
crw-rw---- 1 root disk     10, 237 Mar 31 10:17 loop-control
brw-rw---- 1 root disk      7,   0 Mar 31 10:17 loop0
brw-rw---- 1 root disk      7,   1 Mar 31 10:17 loop1
brw-rw---- 1 root disk      7,   2 Mar 31 10:17 loop2
brw-rw---- 1 root disk      7,   3 Mar 31 10:17 loop3
brw-rw---- 1 root disk      7,   4 Mar 31 10:17 loop4
brw-rw---- 1 root disk      7,   5 Mar 31 10:17 loop5
brw-rw---- 1 root disk      7,   6 Mar 31 10:17 loop6
```

```
semini@Semini-Chamathka:/bin$ cd /home && ls -l
total 4
drwxr-x--- 4 semini semini 4096 Mar 22 22:26 semini
```

```
semini@Semini-Chamathka:/dev$ cd /etc && ls -l
total 796
drwxr-xr-x 2 root root       4096 Aug  5  2025 PackageKit
drwxr-xr-x 7 root root       4096 Aug  5  2025 X11
-rw-r--r-- 1 root root       3444 Jul  5  2023 adduser.conf
drwxr-xr-x 2 root root       4096 Feb 12 08:59 alternatives
drwxr-xr-x 2 root root       4096 Aug  5  2025 apparmor
drwxr-xr-x 9 root root       4096 Mar 28 06:48 apparmor.d
drwxr-xr-x 3 root root       4096 Aug  5  2025 apport
drwxr-xr-x 8 root root       4096 Aug  5  2025 apt
-rw-r--r-- 1 root root       2319 Mar 31  2024 bash.bashrc
-rw-r--r-- 1 root root         45 Jan 25  2020 bash_completion
drwxr-xr-x 2 root root       4096 Aug  5  2025 bash_completion.d
-rw-r--r-- 1 root root        367 Aug  2  2022 bindresvport.blacklist
drwxr-xr-x 2 root root       4096 Apr 19  2024 binfmt.d
drwxr-xr-x 2 root root       4096 Aug  5  2025 byobu
drwxr-xr-x 3 root root       4096 Aug  5  2025 ca-certificates
-rw-r--r-- 1 root root       6288 Aug  5  2025 ca-certificates.conf
drwxr-xr-x 5 root root       4096 Jan 28 10:06 cloud
drwxr-xr-x 2 root root       4096 Aug  5  2025 console-setup
drwx------ 2 root root       4096 Apr 19  2024 credstore
drwx------ 2 root root       4096 Apr 19  2024 credstore.encrypted
drwxr-xr-x 2 root root       4096 Aug  5  2025 cron.d
drwxr-xr-x 2 root root       4096 Feb 12 08:59 cron.daily
drwxr-xr-x 2 root root       4096 Aug  5  2025 cron.hourly
drwxr-xr-x 2 root root       4096 Aug  5  2025 cron.monthly
drwxr-xr-x 2 root root       4096 Aug  5  2025 cron.weekly
drwxr-xr-x 2 root root       4096 Aug  5  2025 cron.yearly
```

```
semini@Semini-Chamathka:/dev$ cd /etc && ls -l
total 796
drwxr-xr-x 2 root root       4096 Aug  5  2025 PackageKit
drwxr-xr-x 7 root root       4096 Aug  5  2025 X11
-rw-r--r-- 1 root root       3444 Jul  5  2023 adduser.conf
drwxr-xr-x 2 root root       4096 Feb 12 08:59 alternatives
drwxr-xr-x 2 root root       4096 Aug  5  2025 apparmor
drwxr-xr-x 9 root root       4096 Mar 28 06:48 apparmor.d
drwxr-xr-x 3 root root       4096 Aug  5  2025 apport
drwxr-xr-x 8 root root       4096 Aug  5  2025 apt
-rw-r--r-- 1 root root       2319 Mar 31  2024 bash.bashrc
-rw-r--r-- 1 root root         45 Jan 25  2020 bash_completion
drwxr-xr-x 2 root root       4096 Aug  5  2025 bash_completion.d
-rw-r--r-- 1 root root        367 Aug  2  2022 bindresvport.blacklist
drwxr-xr-x 2 root root       4096 Apr 19  2024 binfmt.d
drwxr-xr-x 2 root root       4096 Aug  5  2025 byobu
drwxr-xr-x 3 root root       4096 Aug  5  2025 ca-certificates
-rw-r--r-- 1 root root       6288 Aug  5  2025 ca-certificates.conf
drwxr-xr-x 5 root root       4096 Jan 28 10:06 cloud
drwxr-xr-x 2 root root       4096 Aug  5  2025 console-setup
drwx------ 2 root root       4096 Apr 19  2024 credstore
drwx------ 2 root root       4096 Apr 19  2024 credstore.encrypted
drwxr-xr-x 2 root root       4096 Aug  5  2025 cron.d
drwxr-xr-x 2 root root       4096 Feb 12 08:59 cron.daily
drwxr-xr-x 2 root root       4096 Aug  5  2025 cron.hourly
drwxr-xr-x 2 root root       4096 Aug  5  2025 cron.monthly
drwxr-xr-x 2 root root       4096 Aug  5  2025 cron.weekly
drwxr-xr-x 2 root root       4096 Aug  5  2025 cron.yearly
-rw-r--r-- 1 root root       1136 Mar 31  2024 crontab
drwxr-xr-x 4 root root       4096 Aug  5  2025 dbus-1
drwxr-xr-x 3 root root       4096 Aug  5  2025 dconf
-rw-r--r-- 1 root root       2967 Apr 12  2024 debconf.conf
-rw-r--r-- 1 root root         11 Apr 22  2024 debian_version
drwxr-xr-x 2 root root       4096 Aug  5  2025 default
-rw-r--r-- 1 root root       1706 Jul  5  2023 deluser.conf
```

- /bin → system binaries
- /home → user directories
- /dev → device files
- /etc → configuration files
- /lib → libraries

In /dev:

- c → character device
- b → block device

4. Use following methods to go back to your home directory.

cd
cd ~

cd $HOME

Can you distinguish the difference among cd, cd ., cd .. commands?

```
semini@Semini-Chamathka:/lib$ cd
semini@Semini-Chamathka:~$ cd
semini@Semini-Chamathka:~$ cd ~
semini@Semini-Chamathka:~$ cd $HOME
semini@Semini-Chamathka:~$ cd .
semini@Semini-Chamathka:~$ cd ..
semini@Semini-Chamathka:/home$
semini@Semini-Chamathka:/home$ cd
semini@Semini-Chamathka:~$
```

Answer:

- cd → go to home directory
- cd ~ → home directory
- cd $HOME → home directory variable

Difference:

- cd . → current directory (no change)
- cd .. → parent directory

Activity 2

File Access permissions

1. Create a new directory by giving your index number as directory name. Use cd (change directory) command to move into new directory. Now create a new file called chkpermission and check default file permissions using ls -l.

```
semini@Semini-Chamathka:~$ mkdir 24101881
semini@Semini-Chamathka:~$ cd 24101881
semini@Semini-Chamathka:~/24101881$ touch chkpermission
semini@Semini-Chamathka:~/24101881$ ls -l
total 0
-rw-r--r-- 1 semini semini 0 Mar 31 12:48 chkpermission
```

2. Change file permissions of chkpermission by giving full access to owner and remove all permissions from others. Write command(s) executed.

```
semini@Semini-Chamathka:~/24101881$ chmod 700 chkpermission
semini@Semini-Chamathka:~/24101881$
```

3. Change file permission giving full control to owner while others getting read and execute permissions. Write commands and output you are getting.

```
semini@Semini-Chamathka:~/24101881$ chmod 755 chkpermission
ls -l
total 0
-rwxr-xr-x 1 semini semini 0 Mar 31 12:48 chkpermission
semini@Semini-Chamathka:~/24101881$
```

Repeat step 1, 2 and 3, use octal numbers to change permissions. Make sure to create a new file and start working on it.

```
semini@Semini-Chamathka:~/24101881$ touch testfile
chmod 700 testfile
chmod 744 testfile
```

700 – owner full access only
744 – owner full, other read

4. Now go back to parent directory. (It is your home directory). Check directory permissions which are created with your index number. Write down permission available by defaults for directory.

```
semini@Semini-Chamathka:~$ cd ..
ls -l
total 4
drwxr-x--- 5 semini semini 4096 Mar 31 12:53 semini
semini@Semini-Chamathka:/home$
```

5. Now change directory permissions by removing all permissions from all others. Write it.
6. Now remove execute and read permissions from owner. Try to see the contents of the directory. Observe the result.

```
semini@Semini-Chamathka:~/24101881$ chmod u-rx .
semini@Semini-Chamathka:~/24101881$ ls -ld 24101881
ls: cannot access '24101881': Permission denied
```

7. Assign read permission to directory and check to see content of the directory. Ob- serve the result.
   - Permission denied
   - Cannot access directory

8. Add read and execute permissions to the directory and try to see the content. Write your observation in order to see directory contents.

```
semini@Semini-Chamathka:~/24101881$ cd 24101881
ls
-bash: cd: 24101881: Permission denied
ls: cannot open directory '.': Permission denied
semini@Semini-Chamathka:~/24101881$
```

   - Directory contents can be viewed again

Activity 3

Changing file/directory modes

1. Create a directory, lab1 in your home directory. Create a text file, idioms with following text.

   a. Try, Try, Try, One day you can fly
   b. Ideas without action are worthless
   c. What doesn't kill me makes me stronger
   d. Patience is the companion of wisdom
   e. Only those who do nothing make no mistakes

```
semini@Semini-Chamathka:~$ mkdir lab1
cd lab1
semini@Semini-Chamathka:~/lab1$ nano idioms
semini@Semini-Chamathka:~/lab1$ ls -l
total 4
-rw-r--r-- 1 semini semini 190 Mar 31 13:08 idioms
```

```
  GNU nano 7.2                          idioms
Try, Try, Try, One day you can fly
Ideas without action are worthless
What doesn't kill me makes me stronger
Patience is the companion of wisdom
Only those who do nothing make no mistakes
```

2. Check access permissions by typing ls -l. Type chmod a+rwx idioms and check the access permissions again. Type chmod 644 idioms and re-check the access permissions. Practice chmod command in both modes.

```
semini@Semini-Chamathka:~/lab1$ ls -l
total 4
-rw-r--r-- 1 semini semini 190 Mar 31 13:08 idioms
semini@Semini-Chamathka:~/lab1$ chmod a+rwx idioms
semini@Semini-Chamathka:~/lab1$ chmod 644 idioms
semini@Semini-Chamathka:~/lab1$ ls -l
total 4
-rw-r--r-- 1 semini semini 190 Mar 31 13:08 idioms
```

   - 644 → owner read/write, others read only

3. Type **ls -i** to check the i-node number of the idioms file.

```
semini@Semini-Chamathka:~/lab1$ ls -i idioms
2607 idioms
```

4. Check the number of links to idioms file.

```
semini@Semini-Chamathka:~/lab1$ ls -l idioms
-rw-r--r-- 1 semini semini 190 Mar 31 13:08 idioms
```

   - Link count = 1

Activity 4

Hard links and soft links

1. Create a directory, linkdemo in your home directory. Inside that directory, create two directories softln and hardln.

```
semini@Semini-Chamathka:~$ mkdir linkdemo
semini@Semini-Chamathka:~$ cd linkdemo
semini@Semini-Chamathka:~/linkdemo$ mkdir softln hardln
semini@Semini-Chamathka:~/linkdemo$
```

2. Inside hardln directory, create a hard link, hardidioms to idioms file.

```
semini@Semini-Chamathka:~/linkdemo$ cd hardln
semini@Semini-Chamathka:~/linkdemo/hardln$ ln ../../lab1/idioms hardidioms
semini@Semini-Chamathka:~/linkdemo/hardln$
```

3. Check the link count of the idioms file, link count of the hardidioms file, size of the two files, and i-node number of the hardidioms file.

```
semini@Semini-Chamathka:~/linkdemo/hardln$ ls -li ../../lab1/idioms hardidio
ms
2607 -rw-r--r-- 2 semini semini 190 Mar 31 13:08 ../../lab1/idioms
2607 -rw-r--r-- 2 semini semini 190 Mar 31 13:08 hardidioms
```

- Same inode number
- Same size
- Link count increases

4. Edit the idioms file and observe the changes in hardidioms file and vice versa. Check the size of files after each revision.

```
  GNU nano 7.2                    ../../lab1/idioms
Try, Try, Try, One day you can fly
Ideas without action are worthless
What doesn't kill me makes me stronger
Patience is the companion of wisdom
Only those who do nothing make no mistakes
```

- Changes reflect in both files (same inode)

5. Create another hard link and check the following properties of the original file and other hard links.

   Size of the file, number of links and i-node numbers.

   Delete a link file and check the above properties again.

```
semini@Semini-Chamathka:~/linkdemo/hardln$ ln ../../lab1/idioms hardidioms2
semini@Semini-Chamathka:~/linkdemo/hardln$ rm hardidioms
semini@Semini-Chamathka:~/linkdemo/hardln$ ls -li
total 4
2607 -rw-r--r-- 2 semini semini 190 Mar 31 13:22 hardidioms2
```

- File still exists
- Data not deleted until last link removed

6. Inside softln directory, create a soft link, softidioms to idioms file.

```
semini@Semini-Chamathka:~/linkdemo/hardln$ cd ../softln
ln -s ../../lab1/idioms softidioms
```

7. Check the link count of the idioms file, link count of the hardidioms file, link count of the softidioms file, size of the files, and i-node number of the softidioms file.

```
semini@Semini-Chamathka:~/linkdemo/softln$ ls -li ../../lab1/idioms softidioms
2607 -rw-r--r-- 2 semini semini 190 Mar 31 13:22 ../../lab1/idioms
2647 lrwxrwxrwx 1 semini semini  17 Mar 31 13:24 softidioms -> ../../lab1/idioms
```

- Different inode numbers
- Soft link points to original file
- Original link count unchanged

8. Edit the idioms file and observe the changes in softidioms file and vice versa. Check the size of files after each revision.

```
semini@Semini-Chamathka:~/linkdemo/softln$ rm softidioms
semini@Semini-Chamathka:~/linkdemo/softln$ ls -l
total 0
```

- Original file unaffected
- Only shortcut removed

9. Create another soft link from different directory structure of your own and check the following properties of the original file and other soft links.

   - Size of the file, number of links and i-node numbers
   - Delete a soft link file and check the above properties again.

   Write your observations:

```
semini@Semini-Chamathka:~/linkdemo/softln$ cd ~
semini@Semini-Chamathka:~$ mkdir linktest
semini@Semini-Chamathka:~$ cd linktest
semini@Semini-Chamathka:~/linktest$ ^C
semini@Semini-Chamathka:~/linktest$ ln -s ../lab1/idioms softidioms2
semini@Semini-Chamathka:~/linktest$ ls -li ../lab1/idioms softidioms2
2607 -rw-r--r-- 2 semini semini 190 Mar 31 13:22 ../lab1/idioms
2648 lrwxrwxrwx 1 semini semini  14 Mar 31 13:31 softidioms2 -> ../lab1/idioms
semini@Semini-Chamathka:~/linktest$ rm softidioms2
semini@Semini-Chamathka:~/linktest$ ls -li ../lab1/idioms
2607 -rw-r--r-- 2 semini semini 190 Mar 31 13:22 ../lab1/idioms
semini@Semini-Chamathka:~/linktest$
```

Observations:

- The soft link is created in a different directory structure successfully
- The original file and soft link have different inode numbers
- The original file size remains unchanged
- The soft link has a very small size as it stores only the path
- Deleting the soft link does not affect the original file
- Number of links and inode of original file remain unchanged